Monte Carlo

October 30, 2026

Chapter 1

Details

1.1 How is it related to MCWF?

In the MCWF method, the quantum state evolves under a non-Hermitian effective Hamiltonian between stochastic quantum jumps.

My .ipynb notebook implements the non-Hermitian time-evolution part, but it does not implement quantum jumps, stochastic trajectory sampling, or ensemble averaging.

Therefore, it is more accurate to describe this notebook as a preliminary numerical study of effective-Hamiltonian evolution rather than a complete MCWF simulation.

There are also two implementation details worth checking: the short-time calculation references psi before it is initialized in the notebook's execution order, and the time array in the evolution loop has a slight mismatch with the number of Euler steps.

1.2 Numerical Simulation of a Two-Level Quantum System Using an Effective Hamiltonian

This simulation studies the time evolution of a driven two-level quantum system in the presence of decay. A non-Hermitian effective Hamiltonian is used to describe the dynamics, and the time-dependent Schrödinger equation is solved numerically using the Euler method. The resulting state populations are plotted to investigate the system's evolution. This provides a foundation for implementing the Monte Carlo Wave Function method.